

Impact Evaluation Plan

Integrated Dengue Early Warning System (iDEWS)

PREPARED FOR

Philippines Department of Health
Epidemiology Bureau

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Program design is illustrative; epidemiological data drawn from cited primary sources.*

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1 Executive Summary

The Philippines carries one of the highest dengue burdens in Southeast Asia — a burden that official surveillance captures at less than a quarter of its true scale, that falls disproportionately on children aged 5–14, and that is projected to worsen under every climate warming scenario through 2050. The 2017 suspension of the national Dengvaxia school vaccination program, following the manufacturer’s disclosure that the vaccine could worsen outcomes in previously unexposed individuals, effectively closed the vaccine-based prevention pathway. With vaccination off the table as a near-term strategy, early detection and rapid response became the primary lever available to the Philippine government — and the surveillance infrastructure to support that response does not yet exist at the speed and granularity the problem requires.

What the program does. iDEWS — the Integrated Dengue Early Warning System — deploys three components across 150 municipalities and approximately 3,200 barangays in three high-burden regions: a mobile case reporting platform that replaces paper-based weekly submission with real-time digital entry by barangay health workers; a machine learning alert layer that aggregates incoming case data with historical PIDS records and climate inputs to generate weekly outbreak probability scores at the municipality level; and a structured training program for approximately 4,800 barangay health workers and municipal health officers. Implemented by the Philippines DOH Epidemiology Bureau with funding from USAID Philippines (50%), WHO WPRO (30%), and the Government of the Philippines (20%), the program is in its pilot phase (2025–2026), with national scale-up across all 17 administrative regions planned for 2027–2028 pending evaluation findings.

What this evaluation asks. This Impact Evaluation Plan addresses two primary questions: whether iDEWS reduces dengue incidence in target municipalities relative to comparable untreated municipalities, and whether it reduces dengue mortality. Secondary questions examine whether the program improves case reporting completeness and timeliness, whether the ML alert layer generates actionable outbreak warnings ahead of passive surveillance, and whether effects are largest in Geographically Isolated and Disadvantaged Area (GIDA) barangays where the baseline surveillance gap is widest. The learning agenda extends beyond these answerable questions to a broader one: whether community-anchored digital surveillance networks, paired with predictive modeling, can close the gap between disease onset and government response in ways that centralized systems cannot — a finding of direct relevance to WHO WPRO, USAID’s Global Health Security agenda, and the multilateral community investing in pandemic preparedness infrastructure.

Why PSM, and what the proof-of-concept shows. Because iDEWS municipalities were selected on the basis of dengue burden and health system capacity rather than through random assignment, a naive comparison of treated and untreated municipalities would conflate program effects with the pre-existing characteristics that drove selection. Propensity Score Matching addresses this by constructing an artificial control group — matching each treated barangay to an untreated barangay with similar observable characteristics. Using barangay-level data from Project CCHAIN (Epimetrix/HDX, 2024) and Cagayan de Oro City as a proof-of-concept, the analysis confirms the severity of the selection bias problem: before matching, treated barangays averaged 205 annual cases versus 87 for controls — a 135% difference driven entirely by DOH’s burden-based targeting. After caliper matching, the Average Treatment Effect on the Treated falls to 41 cases (136 vs. 95), statistically significant at $p < 0.001$. A pooled two-city robustness analysis (172 barangays, 43 matched pairs) returns a near-identical baseline gap of 40 cases, brings population — the dominant selection axis — into balance, and substantially reduces the wealth imbalance, demonstrating that residual imbalance in the single-city proof-of-concept reflects matching depth rather than a flaw in the design — and that scaling to the full 150-municipality iDEWS dataset will support credible balance.

Theory of change. iDEWS operates through a three-step causal chain: real-time case capture at the barangay level feeds an ML alert layer that detects the statistical signature of an emerging outbreak two to three weeks ahead of passive surveillance, giving municipal health officers time to pre-position resources and activate response before the surge arrives. Earlier detection also compresses the clinical window — patients who present early fare substantially better than those who arrive in crisis. The chain depends on consistent platform adoption by barangay health workers, alert lead time sufficient for a meaningful response, and health officers with the capacity and authority to act on what they receive.

Monitoring. A quarterly monitoring system tracks platform deployment, reporting timeliness and completeness, ML alert lead time and accuracy, and outcome data collection rates against pre-committed review thresholds. Adaptive management triggers are pre-specified: if reporting completeness falls below 80% for two consecutive months, a targeted retraining review is triggered within 30 days; if alert lead time deteriorates, the model is recalibrated before peak dengue season.

Dissemination. The evaluation is built backward from the DOH’s FY 2027–28 budget cycle — the scale-up decision window the findings are designed to inform. An interim report at approximately month 18 delivers early implementation findings; the full evaluation report is delivered by March 2027. Different audiences receive different formats: DOH leadership and USAID receive an executive brief timed to the budget window; Regional Health Offices receive operational implementation findings; WHO WPRO and multilateral partners receive the full evaluation report and working paper.

Contribution beyond dengue. The pilot’s value extends beyond whether it reduces cases in three Philippine regions. The Philippines — with 7,641 islands, a devolved health system, and documented surveillance gaps at the barangay level — is among the most demanding environments in which to test a community-anchored digital surveillance model. If iDEWS works here, it demonstrates that bottom-up digital reporting paired with predictive modeling can generate real-time epidemiological intelligence at the speed and granularity public health response requires. That finding has implications for how governments and multilateral partners invest in early warning infrastructure well beyond dengue, and well beyond the Philippines.

2 Program Description

Dengue surveillance in the Philippines fails not at the national level but at the barangay — where disease first appears, where detection begins, and where the opportunity for early intervention is greatest. The country’s devolved health system places service delivery responsibility with approximately 2,500 Local Government Unit health offices of widely varying capacity across 7,641 islands, making the barangay health worker the critical — and often the weakest — node in the outbreak detection chain. Under the existing passive Philippine Integrated Disease Surveillance and Response–Epidemic-prone Disease Case Surveillance (PIDSR-EDCS) framework, cases are recorded on paper and submitted weekly through a reporting chain that takes two to four weeks to reach the regional level (DOH, n.d.). By the time an outbreak is flagged centrally, it has typically already peaked.

iDEWS was developed in 2024 to address this structural gap. The program is implemented by the Philippines Department of Health Epidemiology Bureau, in coordination with Regional Health Offices and Local Government Units, with USAID Philippines providing 50% of funding, WHO WPRO contributing technical assistance and co-financing (30%), and the Government of the Philippines contributing the remainder. It operates across three regions — CAR, Cagayan Valley (Region II), and Western Visayas (Region VI) — covering approximately 150 municipalities and

3,200 barangays, with an estimated 6.2 million end beneficiaries. The three regions were selected for their consistently high dengue incidence rates and their elevated concentrations of Geographically Isolated and Disadvantaged Area (GIDA) barangays, as designated by the Philippines Department of Health — communities where structural barriers to timely health reporting compound an already severe disease burden. The program is currently in its pilot phase (2025–2026), with national scale-up across all 17 administrative regions planned for 2027–2028 pending evaluation findings.

iDEWS delivers three core components. A mobile-based case reporting platform is deployed to barangay health workers for real-time dengue case submission through the existing PIDS-EDCS infrastructure, replacing paper-based weekly reporting with immediate digital entry at the point of contact. A machine learning alert layer aggregates incoming case reports with historical PIDS data and climate and rainfall inputs to generate weekly outbreak probability scores at the municipality level — converting real-time surveillance data into actionable lead time ahead of passive surveillance. A structured training program equips barangay health workers and municipal health officers with updated WHO 2009 dengue classification protocols and platform use, ensuring the human infrastructure the system depends on is prepared to use it consistently.

3 Evaluation Questions and Learning Agenda

3.1 Primary Evaluation Questions

The evaluation is organized around two primary questions, each corresponding to a core outcome the program is designed to move.

The first is whether iDEWS reduces dengue incidence in target municipalities relative to comparable untreated municipalities. This is the headline impact question — the direct test of whether a barangay-integrated digital surveillance and early warning system, by accelerating case detection and enabling faster outbreak response, produces a measurable reduction in the number of dengue cases reported over the follow-up period. It is answered through the PSM framework described in Section 6, which compares dengue incidence trajectories in treated barangays against a matched control group constructed from municipalities with similar pre-intervention burden, geography, wealth, and health system capacity.

The second primary question is whether iDEWS reduces dengue mortality in target municipalities relative to comparable untreated municipalities. Mortality is the downstream consequence of delayed detection and inadequate outbreak response, and its reduction is the clearest signal that faster surveillance translates into lives saved rather than only administrative improvement. It is treated as a co-primary outcome — analytically parallel to incidence but expected to move more slowly and with greater noise, since case fatality is sensitive to hospital capacity and clinical management factors outside iDEWS’s direct control.

3.2 Secondary Evaluation Questions

The secondary questions examine the mechanisms through which the primary effects are expected to operate, and the populations for whom effects are largest.

Does iDEWS improve dengue case reporting completeness and timeliness at the barangay level? This is the proximal output question — the most direct test of whether the mobile reporting platform is functioning as designed. If iDEWS works through the mechanism its theory of change specifies, reporting rates should rise and reporting lag should fall in treated barangays before any downstream incidence or mortality effects are detectable.

Does the ML alert layer generate outbreak warnings with sufficient lead time for a meaningful public health response? A surveillance system that detects outbreaks after they peak has limited operational value. This question examines whether the algorithm’s weekly municipality-level outbreak probability scores provide actionable lead time — defined as detection at least two weeks prior to peak incidence — relative to the baseline passive surveillance system.

Are effects larger in GIDA barangays, where baseline surveillance capacity was lowest? The theory of change predicts that iDEWS will produce the largest gains precisely where the surveillance gap was widest. Testing this heterogeneity is both a validity check on the mechanism and a targeting question — if GIDA concentration moderates the effect, it informs how a national scale-up should prioritize deployment across the Philippines’ 17 regions.

3.3 Learning Agenda

The evaluation questions above are designed to be answerable within the pilot window. The learning agenda is broader — it frames what the findings mean for decisions that extend well beyond iDEWS itself.

The most immediate decision belongs to the Philippines DOH and its funding partners. If the pilot demonstrates that barangay-integrated digital surveillance reduces dengue incidence and mortality in high-burden regions, DOH and USAID face a scale-up decision: whether to expand iDEWS to all 17 administrative regions in the 2027–2028 phase, and how to prioritize deployment across regions with varying GIDA concentrations and health system capacity. A null result carries its own information — if incidence and mortality do not improve despite improved reporting completeness, it suggests the binding constraint is not surveillance but response capacity, redirecting investment accordingly.

The deeper question the pilot is positioned to answer is whether bottom-up digital surveillance networks can generate real-time epidemiological intelligence at the speed and granularity public health response actually requires. The Philippines, with its 7,641 islands, devolved health system, and documented surveillance gaps, is among the most demanding environments in which to test this model. If iDEWS works here — integrating barangay health workers, municipal health officers, and a machine learning alert layer into a functioning early warning chain — it demonstrates something with implications well beyond dengue: that community-anchored digital reporting, paired with predictive modeling, can close the gap between disease onset and government response in ways that centralized surveillance systems cannot. That finding is of direct interest to WHO WPRO, USAID’s Global Health Security agenda, and the broader multilateral community investing in pandemic preparedness infrastructure. The pilot’s contribution is not only whether it reduces dengue cases in three Philippine regions, but whether it proves a model that could reshape how the world detects and responds to emerging infectious disease threats.

3.4 Evaluation Type: Impact, Performance, or Hybrid

This plan is a hybrid evaluation weighted toward impact. The impact component is the quasi-experimental design in Section 6: Propensity Score Matching, and prospectively a difference-in-differences estimator layered on the matched sample, is what allows the evaluation to make causal claims about whether iDEWS reduces dengue incidence and mortality rather than merely whether treated and untreated municipalities differ. The performance component is the monitoring system in Section 7, which tracks whether the program was delivered as designed — platform deployment, reporting completeness, alert lead time and accuracy. The two are not separable. Because iDEWS works through a multi-step chain, an apparent absence of impact is interpretable only against the

implementation record: a null incidence result paired with strong reporting completeness points to a response-capacity constraint, while a null result paired with weak platform adoption points to an implementation failure. The performance evaluation is therefore a precondition for interpreting the impact evaluation, not a parallel exercise.

4 Theory of Change, Logic Model, and Indicators

4.1 Theory of Change

Problem Statement. The Philippines’ dengue surveillance system fails at the barangay — the administrative unit where disease first appears and where the opportunity for early intervention is greatest. Under the current passive Philippine Integrated Disease Surveillance and Response (PIDSR) framework, barangay health workers record cases on paper and submit weekly reports through a chain that takes two to four weeks to reach the regional level. By the time an outbreak is flagged centrally, it has typically already peaked. The result is a surveillance system that is structurally incapable of generating the lead time public health response requires. As Undurraga et al. (2017) documented, the Philippines’ passive surveillance captures approximately 21% of true dengue burden — a gap that reflects not only underreporting but the structural lag between case onset and administrative detection. In high-burden, geographically isolated barangays, that lag is longest and its consequences most severe.

Mechanism of Change. iDEWS addresses this problem through a three-step causal chain.

The first step is real-time case capture. The mobile reporting platform replaces paper-based weekly submission with immediate digital case entry at the point of contact, collapsing the reporting lag from weeks to hours. This is the necessary precondition for everything downstream — without timely case data flowing from the barangay level, neither the alert layer nor the response chain can function as designed.

The second step is outbreak prediction. The ML alert layer aggregates incoming case reports with historical PIDSR incidence data and climate and rainfall inputs to generate weekly outbreak probability scores at the municipality level. By detecting the statistical signature of an emerging outbreak before case counts reach the threshold that would trigger a manual alert, the algorithm converts real-time surveillance data into actionable lead time — estimated at two to three weeks ahead of passive surveillance under the program model.

The third step is accelerated response. Municipal health officers receiving a high-probability outbreak alert can pre-position supplies, activate vector control measures, issue community advisories, and prepare clinical capacity before the surge arrives. Earlier detection also compresses the clinical window — patients who present early in the disease course, before dengue progresses to severe hemorrhagic disease, have substantially better outcomes. The downstream result, if the chain holds, is fewer total cases through contained transmission and lower case fatality through earlier clinical presentation.

Key Assumptions. For this mechanism to produce the expected outcomes, the following assumptions must hold:

- Barangay health workers consistently use the mobile platform and submit cases in real time rather than reverting to paper-based workflows. If adoption is inconsistent — particularly in GIDA barangays with limited connectivity — the real-time data stream the ML layer depends on will be incomplete.
- The ML alert layer generates outbreak warnings with sufficient lead time for a meaningful public

health response. A system that detects outbreaks at or after peak has no operational value regardless of technical accuracy.

- Municipal health officers act on alerts. Early warning is necessary but not sufficient — response capacity, supply chain reliability, and local political will determine whether a warning translates into a mobilized response.
- Improved detection leads to improved clinical outcomes. This assumption is most vulnerable in settings where hospital capacity is the binding constraint rather than detection timing — if facilities are overwhelmed regardless of warning, earlier detection may not reduce case fatality.

Moderators of Effect. The average effect will mask variation across barangays and regions, and the evaluation is designed to examine three moderators in particular. GIDA status is the most theoretically important — the theory predicts the largest gains in barangays where the baseline surveillance gap was widest, making GIDA concentration the primary heterogeneity test. Connectivity and device access will moderate the adoption step — barangays with reliable mobile infrastructure are more likely to sustain real-time reporting than those dependent on intermittent signal. And baseline health system capacity at the municipal level will moderate the response step — the alert layer can only convert lead time into lives saved where health officers have the resources and authority to act on what they receive.

4.2 Logic Model: The Results Chain

The results chain translates the theory of change into the program’s operational structure, from the resources it consumes through to the population-level outcomes it is ultimately about. The first three levels — inputs, activities, and outputs — are the supply side the implementing organizations control and the monitoring plan in Section 7 verifies. The final two levels are the demand side: participant and population responses that the impact evaluation in Section 6 must establish causally.

4.3 Indicator Matrix

Table 1: Indicator matrix for the iDEWS impact evaluation.

Indicator	Definition	Data Source	Frequency
Platform deployment rate	% of target barangays with active mobile reporting platform	DOH program records	Once at inception
BHWs trained	# of barangay health workers trained as % of target (4,800)	Training logs	Quarterly
Case reporting timeliness	% of cases submitted within 24 hours of diagnosis	PIDSR-EDCS platform logs	Weekly
Reporting completeness	% of expected weekly reports submitted by treated barangays	PIDSR-EDCS platform logs	Weekly
Alert lead time	Days between ML outbreak alert and peak incidence in flagged municipalities	ML alert logs vs. PIDSR	Per outbreak event
Alert accuracy	% of ML alerts followed by confirmed outbreak within 3 weeks	ML alert logs vs. PIDSR	Per outbreak event

Indicator	Definition	Data Source	Frequency
Dengue incidence rate	Confirmed cases per 100,000 population in treated vs. matched controls	PIDSR-EDCS; Project CCHAIN	12 and 24 months
Dengue mortality rate	Dengue deaths per 100,000 population in treated vs. matched controls	PIDSR-EDCS; DOH vital statistics	12 and 24 months
Case fatality rate	Dengue deaths as % of confirmed cases, treated vs. matched controls	PIDSR-EDCS	12 and 24 months
Durable incidence reduction	Sustained incidence reduction beyond 24-month evaluation window	PIDSR-EDCS post-evaluation	Long-term

5 Literature Review

5.1 Program Evidence Base: Dengue Burden, Surveillance Gaps, and the Case for Early Warning in the Philippines

The Philippines carries one of the highest dengue burdens in Southeast Asia, and the scale of that burden is substantially larger than official surveillance figures suggest. Shepard, Undurraga, and Halasa (2015) estimated an annual average of 842,867 clinically diagnosed dengue cases during 2008–2012 — more than seven times the 117,065 cases reported through passive surveillance over the same period — with direct medical costs of \$345 million annually in 2012 USD. A companion study by Undurraga et al. (2017) estimated 794,255 annual dengue episodes for 2010–2014 using a DALY-based burden framework, with a reporting rate of approximately 21%, equivalent to an expansion factor of 4.7. Together these studies establish that the Philippines’ passive surveillance system captures less than a quarter of true dengue burden, making the gap between reported and actual caseload one of the most consequential features of the epidemiological landscape iDEWS is designed to address.

The burden is not evenly distributed. Children aged 5–14 carry the highest national incidence rate, and over 80% of dengue-related deaths occur in pediatric patients (DOH Epidemiology Bureau, 2023). Geographically, CAR recorded the highest dengue incidence rate nationally in both 2022 and 2023 (DOH Epidemiology Bureau, 2023), driven in part by its high concentration of Geographically Isolated and Disadvantaged Area (GIDA) barangays — communities where structural barriers to timely health reporting compound an already elevated disease burden. The surveillance breakdown occurs not at the national level but at the barangay, where most diagnoses are made on clinical grounds without laboratory confirmation and where the Philippines’ devolved health system places reporting responsibility on local government units of widely varying capacity.

The closure of the vaccine-based prevention pathway sharpens the urgency of surveillance investment. In November 2017, Sanofi Pasteur disclosed that Dengvaxia — the dengue vaccine the Philippines had deployed in a national school-based program beginning in April 2016, vaccinating approximately 733,713 children at a cost of PHP 3.5 billion — could worsen dengue outcomes in seronegative individuals, those with no prior dengue exposure (DOH, 2017). The DOH suspended the program immediately. The resulting collapse in public confidence extended well beyond dengue: a Vaccine Confidence Project survey found that the share of Filipinos who strongly agreed vaccines are safe fell from 82% in 2015 to 21% in 2018, a decline its authors tied directly to the Dengvaxia scare (Larson, Hartigan-Go, and de Figueiredo, 2019). Measles immunization coverage fell over the same period, contributing to a major measles outbreak in 2019 (Larson, Hartigan-Go, and

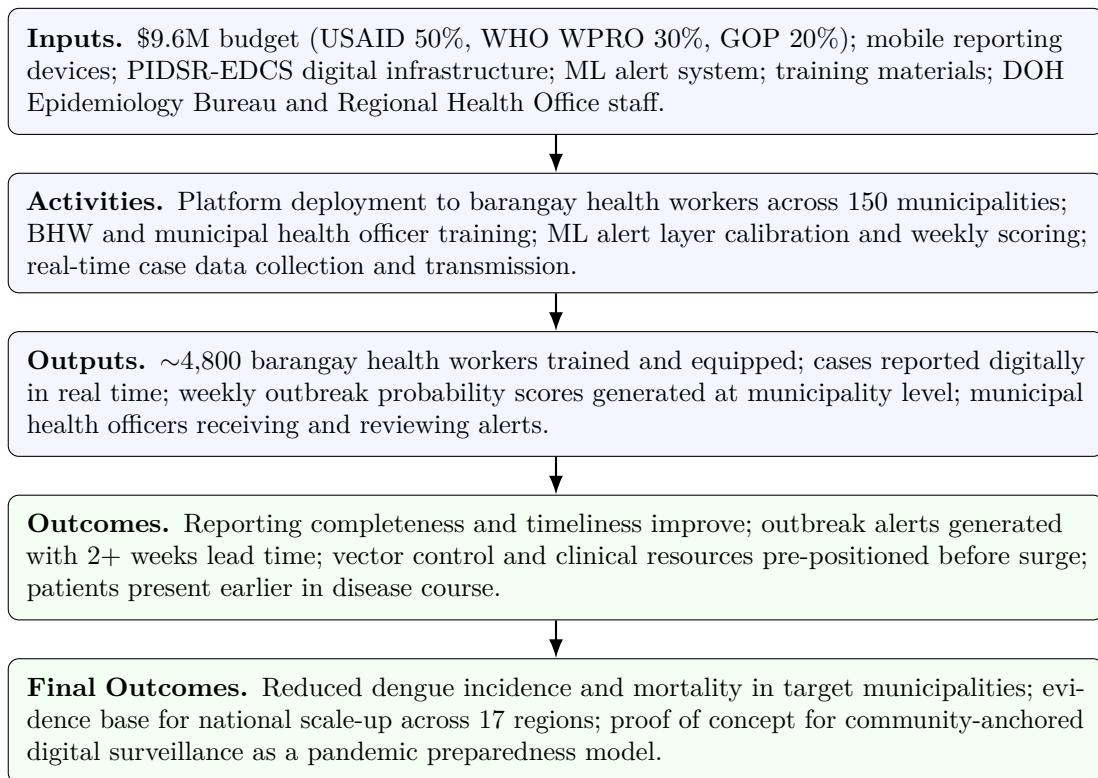


Figure 1: The iDEWS results chain. The break between Outputs and Outcomes is the supply-side / demand-side boundary: levels above it the program controls and monitoring verifies; levels below depend on response and are established through the matched-comparison design.

de Figueiredo, 2019). With vaccination effectively removed as a near-term prevention tool, early detection and rapid response became the primary intervention lever available to the Philippine government — which is the strategic context from which iDEWS emerges.

The urgency is compounded by the climate trajectory. Seposo et al. (2024) project that dengue burden in the Philippines will increase under all warming scenarios through 2050, with the steepest rises concentrated in already-vulnerable regions. By October 2024, reported dengue cases had reached 269,467 — an 82% increase over the same period in 2023 — prompting the DOH Secretary to declare outbreak levels (DOH Epidemiology Bureau, 2024; WHO, 2024). The convergence of a rising baseline burden, a constrained prevention toolkit, and a favorable multilateral investment window in digital public health infrastructure makes the case for a real-time, barangay-integrated surveillance system both urgent and timely.

5.2 Methods Precedent: ML-Based Early Warning and PSM as an Evaluation Design

The use of machine learning for dengue outbreak prediction has expanded rapidly across Southeast Asia, providing both methodological precedent and design benchmarks for iDEWS. Rahman et al. (2025) developed a tree-based ML early warning system for dengue in Bangladesh using Random Forest, XGBoost, and LightGBM models trained on socioeconomic, landscape, and climate inputs, demonstrating that ensemble ML models can generate accurate district-level outbreak predictions with interpretable output suitable for public health decision-making. The iDEWS ML alert layer draws on a comparable modeling approach — aggregating PIDS case reports, historical incidence, and climate and rainfall inputs to generate weekly outbreak probability scores at the municipality level — and the Bangladesh precedent establishes that this architecture is both technically feasible and operationally deployable in a low-resource, high-burden Southeast Asian context.

Because iDEWS municipalities were selected on the basis of dengue burden and health system capacity rather than through random assignment, a naive comparison of treated and untreated municipalities would conflate program effects with the pre-existing characteristics that drove selection. Propensity Score Matching (PSM) addresses this problem by constructing an artificial control group — matching each treated unit to an untreated unit with similar observable characteristics — so that the comparison reflects program effect rather than selection bias. The method originates with Rosenbaum and Rubin (1983), who show that conditioning on the scalar propensity score is sufficient to remove the bias associated with a set of observed covariates, and it has since been developed into a standard toolkit for observational program evaluation (Caliendo and Kopeinig, 2008; Stuart, 2010). It is particularly appropriate for programs deployed through government targeting criteria, where randomization is infeasible but administrative data on potential confounders is available. Its central limitation is equally well established: matching balances only observed covariates, so unlike randomization it cannot account for selection on unobservables, and credible application requires both transparent balance diagnostics and explicit acknowledgment of that residual threat (Heckman, Ichimura, and Todd, 1997). The framework applied in this evaluation matches barangays on six covariates — barangay land area, wealth index, flood hazard exposure, healthcare access, population size, and average rainfall — each selected because theory and prior literature suggest they predict both program selection and dengue outcomes. The Project CCHAIN dataset (Epimetrics/HDX, 2024), which links DOH-PIDS disease records with barangay-level climate, wealth, hazard, and geographic data across 12 Philippine cities, provides the data infrastructure that makes this matching feasible at the barangay level.

6 Evaluation Design

6.1 Design Choice and Rationale

The central challenge in evaluating iDEWS is that the program was not deployed randomly. The Philippines DOH selected the 150 pilot municipalities on the basis of high dengue burden, elevated GIDA barangay concentration, and existing health system capacity — characteristics that also independently predict dengue outcomes. A simple before-and-after comparison, or a naive comparison of treated and untreated municipalities, would confuse the program’s effect with the pre-existing differences that drove selection in the first place. The treated municipalities were already the sickest. Any apparent improvement could reflect regression to the mean, seasonal patterns, or concurrent interventions rather than iDEWS itself.

Randomization would solve this problem cleanly, but it was not feasible here. DOH selected the pilot regions through a technical working group process before this evaluation was commissioned, and withholding a proven surveillance upgrade from high-burden communities for experimental purposes would have been neither ethical nor politically viable.

Propensity Score Matching is the appropriate alternative. The core idea is straightforward: for each barangay that received iDEWS, find a barangay that did not but looks as similar as possible across every observable characteristic that predicted selection. By constructing this artificial control group, PSM makes the comparison fair — the difference in outcomes between treated and matched control barangays can be attributed to the program rather than to pre-existing differences between them.

6.2 Identification Strategy

The quantity of interest is the Average Treatment Effect on the Treated (ATT). For barangay i , let $Y_i(1)$ and $Y_i(0)$ denote dengue outcomes with and without iDEWS, and let $D_i \in \{0, 1\}$ indicate treatment. The ATT is

$$\text{ATT} = E[Y_i(1) - Y_i(0) \mid D_i = 1],$$

the average effect of the program on the barangays that actually received it. The fundamental problem is that $Y_i(0)$ is never observed for treated barangays; matching constructs it from observably similar controls.

Identification under PSM rests on two assumptions. The first is conditional independence, or selection on observables: conditional on the covariate vector X_i , treatment assignment is independent of potential outcomes,

$$(Y_i(0), Y_i(1)) \perp D_i \mid X_i.$$

The second is common support (overlap): every treated barangay has a positive probability of having an untreated counterpart with similar covariates, $0 < \Pr(D_i = 1 \mid X_i) < 1$. Rosenbaum and Rubin (1983) establish that conditioning on the scalar propensity score $e(X_i) = \Pr(D_i = 1 \mid X_i)$ is sufficient to satisfy the first assumption, which is what makes matching on a single estimated score, rather than on the full covariate vector, valid.

These assumptions are demanding, and conditional independence in particular is untestable. For that reason the evaluation’s primary prospective estimand is not cross-sectional matching alone but a difference-in-differences estimator layered on the matched sample. With pre- and post-deployment outcomes, the matched-sample DiD is

$$\widehat{\text{ATT}}_{\text{DiD}} = (\bar{Y}_T^{\text{post}} - \bar{Y}_T^{\text{pre}}) - (\bar{Y}_C^{\text{post}} - \bar{Y}_C^{\text{pre}}),$$

computed over matched treated–control pairs. This relaxes the conditional independence assumption to a weaker conditional parallel-trends assumption — that matched treated and control barangays would have followed the same dengue trajectory absent iDEWS — and differences out any time-invariant unobserved heterogeneity that matching on observables cannot remove. It is the more credible of the two and is the design the full evaluation will run once Phase 2 outcome data exists.

The analysis presented in the remainder of this section is the cross-sectional PSM proof-of-concept. Because iDEWS has not yet deployed, there is no post-period and no DiD to estimate; the proof-of-concept instead demonstrates that the matching machinery is implementable on barangay-level data and quantifies the magnitude of the selection problem the prospective design must overcome.

6.3 Data and Geographic Scope

The PSM analysis uses barangay-level data from Project CCHAIN (Epimetrics/HDX, 2024), a validated open-source dataset linking DOH-PIDSR disease surveillance records with climate, wealth, flood hazard, healthcare access, and geographic data across 12 Philippine cities. Because CCHAIN’s disease data does not cover the three iDEWS target regions directly, the primary analysis uses Cagayan de Oro City (Region X) as a proof-of-concept, with 79 barangays providing the matching pool. Cagayan de Oro is among the highest-burden dengue cities in the CCHAIN dataset and shares key structural characteristics with the iDEWS target regions — high dengue incidence, a devolved health system, and significant GIDA barangay concentration. A robustness analysis (Section 6.8) pools Cagayan de Oro with a second CCHAIN city to widen the matching pool. The methodology is designed to scale to the full 150-municipality iDEWS dataset once equivalent barangay-level data becomes available for CAR, Cagayan Valley, and Western Visayas.

Since iDEWS has not yet been fully deployed, treatment is simulated by designating the top 40% of barangays by average annual dengue case burden as treated — reflecting DOH’s stated targeting logic of prioritizing high-burden areas for the pilot.

6.4 Matching Variables

Six covariates were selected for the propensity score model, each chosen because prior literature and program theory suggest it predicts both program selection and dengue outcomes:

- **Relative wealth index** — poorer barangays have less vector control capacity and were more likely to be prioritized.
- **Flood hazard exposure** — standing water drives *Aedes aegypti* breeding and elevates both burden and targeting priority.
- **Healthcare access** — measured as the proportion of the barangay population within five minutes of a health center, affecting both reporting capacity and treatment-seeking.
- **Population size** — larger barangays generate higher raw case counts and were more likely to meet DOH selection thresholds.
- **Average rainfall** — directly tied to mosquito breeding cycles and a key input to the iDEWS ML alert layer.
- **Barangay land area** — a scale control that, together with population, proxies settlement density; denser barangays sustain more intense *Aedes* transmission, and area conditions how raw case counts map to underlying incidence.

6.5 Propensity Score Estimation and Matching

Propensity scores — each barangay’s estimated probability of being selected into iDEWS — were estimated using logistic regression on the six standardized covariates above. The score distribution before matching shows strong separation between groups: control barangays cluster around a mean propensity score of 0.21, while treated barangays cluster around 0.69, confirming that the two groups are systematically different before matching and that a naive comparison would be severely biased (Figure 2).

Matching was performed using nearest-neighbor caliper matching, which pairs each treated barangay with the control barangay whose propensity score is closest, subject to a maximum distance constraint of 0.2 standard deviations. The caliper prevents poor matches — treated barangays with no sufficiently similar control are dropped rather than matched badly. This trades sample size for match quality, which is the appropriate tradeoff for a small dataset.

Of 32 treated barangays, 10 matched pairs were retained within the caliper. The remaining 22 treated barangays fell outside the common support region — meaning no sufficiently similar control barangay existed in the dataset — and were excluded from the analysis.

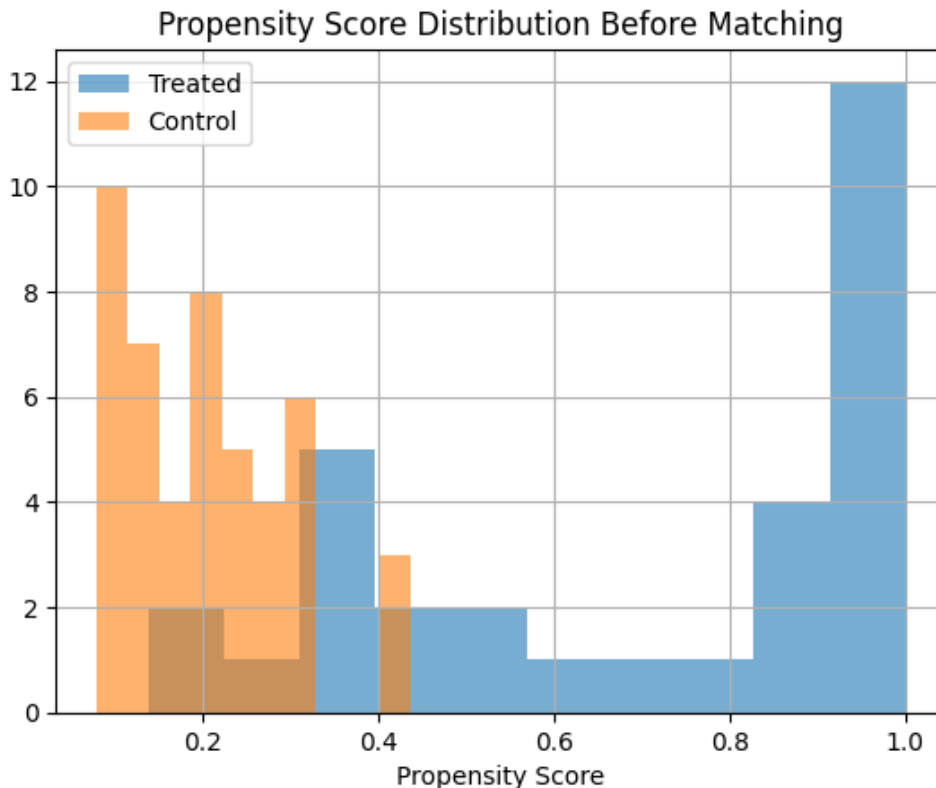


Figure 2: Propensity score distribution before matching. Control barangays (orange) cluster around 0.21; treated barangays (blue) cluster around 0.69, confirming systematic pre-matching differences.

6.6 Balance Assessment

After matching, covariate balance was assessed using Standardized Mean Differences (SMDs), which measure the remaining gap between treated and matched control groups on each covariate. An SMD below 0.1 indicates good balance — the two groups are essentially comparable on that dimension.

The result, shown in the Love plot in Figure 3, is partial and instructive. Matching brings population size — the covariate on which the two groups differed most before matching — into near-perfect balance, collapsing its SMD from roughly 1.2 to near zero. But every other covariate remains above the 0.1 threshold after matching, and on most of them — wealth, flood hazard, rainfall, and land area — the matched groups are markedly less comparable than the unmatched groups, with post-matching SMDs between roughly 0.52 and 0.83.

This pattern is diagnostic rather than disqualifying. With only 10 pairs drawn from 79 barangays, the estimated propensity score is dominated by the single axis on which selection runs most strongly — population scale, the mechanical correlate of the burden-based targeting rule. Matching therefore stabilizes that dominant axis at the expense of the others. This is a known small-sample failure mode of propensity score matching: when one covariate dominates the score, balancing it can disturb the rest. It is best read as evidence that the matching pool is too thin, not that the design is unsound — a reading the pooled robustness analysis in Section 6.8 tests directly.

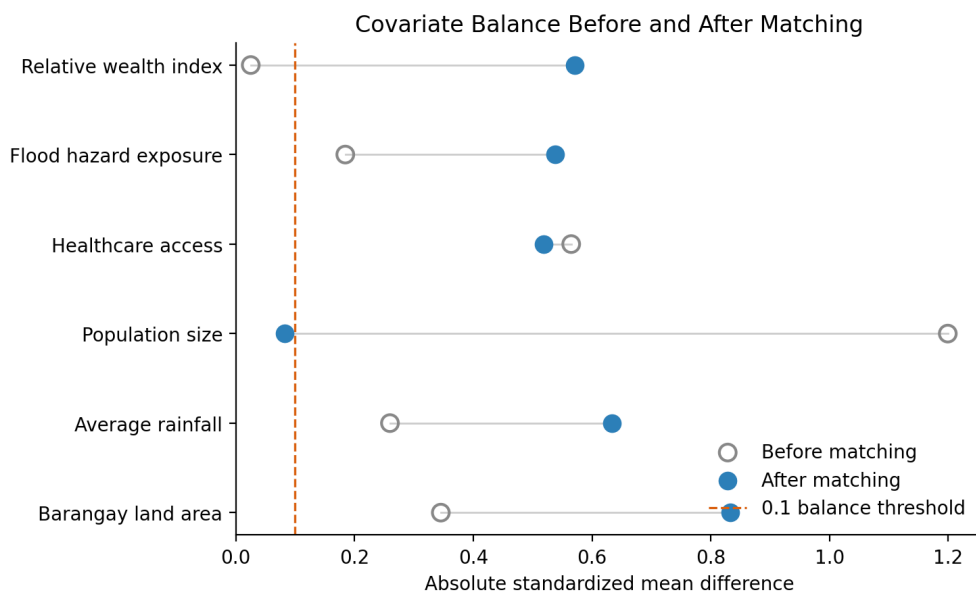


Figure 3: Covariate balance before and after matching, Cagayan de Oro (79 barangays, 10 matched pairs, six covariates). Population balances well; residual imbalance remains on the other covariates. The dashed line marks the 0.1 balance threshold.

6.7 Results

Before matching, treated barangays averaged 205 annual dengue cases compared to 87 for control barangays — a 135% difference driven entirely by DOH’s burden-based targeting, not any program effect. After caliper matching, the Average Treatment Effect on the Treated — the difference in average annual cases between matched treated and control barangays — was 41 cases (136 vs. 95), statistically significant at $p < 0.001$ (Figure 4).

This figure should be interpreted carefully. Because iDEWS has not yet been deployed, it represents the baseline burden difference between targeted and matched control barangays before any intervention — not a program impact estimate. In a prospective evaluation, the difference-in-differences framework of Section 6.2 applied to post-deployment outcome data would isolate the causal effect of iDEWS on dengue incidence and mortality. The significant pre-treatment difference,

even after matching, underscores both the severity of the selection bias problem and the importance of the matched design: without matching, any post-deployment comparison would overstate the program’s effect by conflating it with the pre-existing burden gap. Notably, the pooled robustness analysis below returns a near-identical baseline gap of 40 cases from an independent second city, suggesting the roughly 40-case targeting differential is a stable feature of burden-based selection rather than an artifact of the Cagayan de Oro sample.

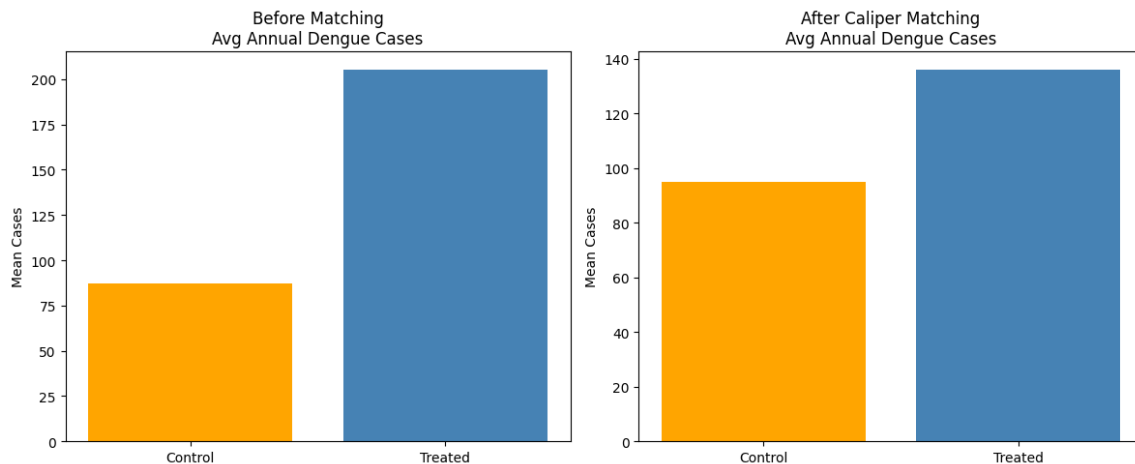


Figure 4: Average annual dengue cases before and after caliper matching, Cagayan de Oro. Matching reduces the treated–control gap from 118 cases to 41 cases (ATT = 41, $p < 0.001$).

6.8 Robustness: Pooled Multi-City Sample

The balance limitation in the single-city proof-of-concept points to matching depth as the binding constraint. To test that diagnosis, the analysis was re-run on a pooled sample combining Cagayan de Oro with Zamboanga City, the two CCHAIN cities with sufficient barangay-level dengue coverage, for a pooled pool of 172 barangays. Two design features make pooling defensible rather than a simple enlargement of N . Treatment is assigned *within* each city — the top 40% by burden in each city, rather than pooled — so that the comparison does not collapse into selecting the higher-incidence city. And the propensity model includes city fixed effects, so the estimated score reflects within-city selection rather than between-city differences in baseline incidence. The pooled model uses the five substantive covariates and drops the barangay land-area scale control, so the robustness check varies both the matching pool and the covariate set.

The pooled analysis retains 43 matched pairs, more than four times the single-city count, and the balance result improves materially (Figure 5). Population size, the dominant axis, balances to an SMD near 0.01; the relative wealth index improves from roughly 0.45 to 0.21; and healthcare access remains well balanced. Residual imbalance persists on flood hazard exposure, which worsens to roughly 0.35, and rainfall is essentially unchanged. The ATT on baseline annual cases is 39.9 cases (treated 144.1 vs. control 104.2, $p < 0.001$), statistically indistinguishable from the 41-case estimate in Cagayan de Oro.

Two findings follow, and both strengthen rather than weaken the design. First, quadrupling the matched pool brings population — the dominant selection axis — into balance and sharply reduces the wealth imbalance, confirming that the imbalance in the single-city proof-of-concept reflects matching depth and not a structural flaw in the approach. By demonstration rather than assertion, this is the evidence that the full 150-municipality iDEWS dataset, far larger than either

pool here, will support credible balance. Second, the near-identical ATT across two independent city pools indicates the baseline burden differential is a stable feature of burden-based targeting.

Two cautions are stated plainly. Balance is improved but not complete: flood hazard exposure remains above threshold after pooling, so the pooled sample is more comparable on the dimensions that matter most for selection but is not uniformly balanced. And Zamboanga, like Cagayan de Oro, is a CCHAIN proxy city rather than one of the three iDEWS target regions — pooling buys statistical depth, not geographic proximity to CAR, Cagayan Valley, or Western Visayas. The within-city assignment and city fixed effects are what make the pooled comparison methodologically sound; they do not make Zamboanga a substitute for the target regions, which only target-region PIDS data can supply.

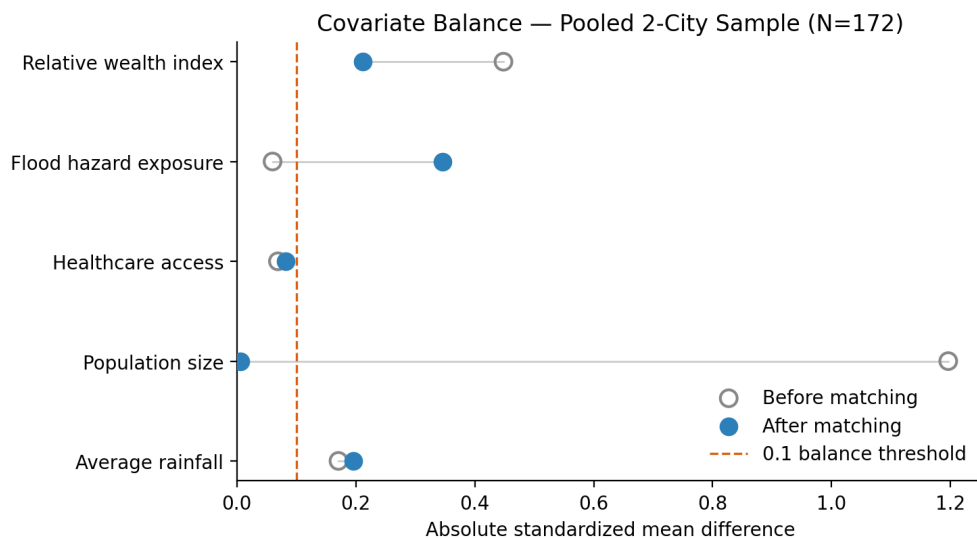


Figure 5: Covariate balance before and after matching, pooled two-city sample (172 barangays, 43 matched pairs, five covariates). Population and wealth come into balance; residual imbalance remains on flood hazard exposure. The dashed line marks the 0.1 balance threshold.

6.9 Threats to Validity

A quasi-experimental design must name the threats it is exposed to and say how each is managed. Five bear on this evaluation.

Selection on unobservables. This is the central limitation of matching relative to randomization: PSM balances only the covariates it observes, so any unmeasured determinant of both iDEWS selection and dengue outcomes — local political will, undocumented vector-control activity, health-officer quality — remains a potential confounder. Two features address it. The prospective difference-in-differences specification (Section 6.2) differences out any time-invariant unobserved heterogeneity, which is the largest part of this threat. For what remains, the full evaluation will report a Rosenbaum sensitivity analysis (Rosenbaum, 2002) quantifying how strong an unobserved confounder would have to be to overturn the estimated effect, so the result’s robustness to hidden bias is stated rather than assumed.

Differential reporting. This threat is specific to iDEWS and is the most important one to confront, because the program is designed to increase case capture. If treated barangays report a larger share of true cases after deployment, measured incidence could rise mechanically even as true transmission falls — biasing the headline incidence estimate toward zero or the wrong sign.

The evaluation manages this in three ways: it treats mortality and case fatality, which are far less sensitive to reporting completeness than incidence, as co-primary outcomes; it carries reporting completeness as a monitored indicator so that incidence movements can be interpreted against changes in capture; and it frames a rise in reported incidence alongside improved completeness as consistent with the theory rather than contradicting it. An incidence estimate read in isolation from the completeness data would be uninterpretable, which is why the hybrid design ties the two together.

Common support and external validity. The caliper drops treated barangays with no comparable control — 22 of 32 in the proof-of-concept. This protects internal validity but narrows the estimate to the matchable middle of the distribution, so the effect may not generalize to the highest-burden barangays that have no control analogue. Pooling widens common support, as the robustness analysis shows, and the full multi-region dataset will widen it further; the limitation is reported transparently rather than hidden in the retained-pair count.

Spillovers. Vector-control surges and community advisories triggered by an alert in a treated barangay could reduce transmission in an adjacent control barangay, contaminating the counterfactual and biasing the estimate toward zero. Where geography permits, matching favors non-adjacent controls, and any contamination of this kind makes the estimate conservative rather than inflated. The evaluation is not designed to estimate the magnitude of spillovers and notes this as a limitation rather than claiming they are absent.

Concurrent interventions. Other dengue programs operating during the evaluation window — national vector-control campaigns, clinical guideline changes — could move outcomes independently of iDEWS. The difference-in-differences design differences out any such shock common to treated and control barangays; the evaluation documents concurrent programs so that any treated-specific co-intervention can be flagged.

6.10 Limitations and Future Directions

Three limitations of the proof-of-concept analysis are worth stating plainly, addressing honestly how imperfect balance may affect findings.

First, the matching pool is small. With 79 barangays in Cagayan de Oro, only 10 matched pairs could be retained within the caliper, and post-matching SMDs remained above the conventional balance threshold. The pooled two-city analysis directly mitigates this, recovering 43 pairs and bringing the dominant covariates into balance, and a larger dataset — covering the full 150 iDEWS municipalities across three regions — would improve both the number of matched pairs and the quality of balance further, producing more precise and trustworthy estimates.

Second, the geographic scope is a proxy. Cagayan de Oro (Region X), and Zamboanga in the pooled analysis, are used as proofs of concept because they hold the richest barangay-level dengue data in the Project CCHAIN dataset. Neither is one of the three iDEWS target regions, and findings may not transfer directly to CAR, Cagayan Valley, or Western Visayas, which differ in health system capacity, GIDA concentration, and dengue seasonality. Municipality-level PIDSR data for the actual target regions would be needed for the full evaluation.

Third, treatment is simulated. The analysis assigns treatment based on baseline burden rather than actual iDEWS deployment records. Once Phase 2 rollout is complete, real treatment indicators will allow the framework to be applied to actual program data, producing a genuine impact estimate rather than a proof-of-concept demonstration.

The intended next step is the difference-in-differences specification set out in Section 6.2, layered on the matched sample — comparing changes in dengue incidence before and after iDEWS deployment in treated barangays against the same change in matched controls. This controls for

any remaining time-invariant unobservables that matching alone cannot address, producing a more credible causal estimate than matching alone.

7 Monitoring Plan

7.1 Monitoring Matrix

The monitoring plan tracks whether iDEWS is being delivered as designed across the pilot period. It focuses on the supply-side levels of the results chain — the inputs, activities, and outputs the implementing organizations control — and attaches to each a review threshold that, if missed, triggers a pre-committed response. Outcome indicators are measured through the matched-comparison framework in Section 6 and are not repeated here.

Table 2: Monitoring matrix for iDEWS.

Indicator	Frequency	Source	Review Threshold
Platform deployment rate	Once at inception, verified quarterly	DOH program records	100% of target barangays active before enrollment opens
BHWs trained	Quarterly	Training logs	$\geq 90\%$ of target (4,800) trained by end of pilot Phase 1
Case reporting timeliness	Weekly	PIDSR-EDCS platform logs	$\geq 80\%$ of cases submitted within 24 hours
Reporting completeness	Weekly	PIDSR-EDCS platform logs	$\geq 80\%$ of expected weekly reports submitted
Alert lead time	Per outbreak event	ML alert logs vs. PIDSR	≥ 2 weeks lead time in $\geq 70\%$ of flagged outbreak events
Alert accuracy	Per outbreak event	ML alert logs vs. PIDSR	$\geq 70\%$ of alerts followed by confirmed outbreak within 3 weeks
Survey/data collection completion	At 12 and 24 months	Evaluation team records	$\geq 80\%$ of target barangays with complete outcome data

7.2 Data Collection Procedures

Monitoring data flows from three streams. Program administrative data — deployment records, training logs, and platform usage statistics — is generated as a byproduct of running the program and compiled into quarterly monitoring reports by the DOH Epidemiology Bureau. PIDSR-EDCS platform logs supply the real-time reporting timeliness and completeness indicators automatically, since the platform records submission timestamps at the point of entry. ML alert logs are maintained by the system and reconciled against PIDSR incidence data at the end of each outbreak event to assess lead time and accuracy. The evaluation team is responsible for outcome data collection at 12 and 24 months, using PIDSR-EDCS records linked to the Project CCHAIN barangay identifiers established in the matched-comparison framework.

7.3 Adaptive Management Triggers

Three triggers are pre-committed. If reporting completeness falls below 80% for two consecutive months in any region, the DOH Epidemiology Bureau and Regional Health Office conduct a targeted retraining and troubleshooting review within 30 days. If alert lead time falls below two weeks in more than 30% of outbreak events in the first six months, the ML alert layer parameters are reviewed and recalibrated before the peak dengue season. If outcome data collection falls below 80% at the 12-month wave, the evaluation team intensifies administrative data linkage through PIDSR-EDCS to compensate for gaps in direct measurement.

8 Learning and Dissemination

8.1 Learning Agenda Revisited

The dissemination plan is built backward from the decisions the evidence is meant to inform. The most time-sensitive decision belongs to the Philippines DOH and its funding partners — whether to proceed with national scale-up across all 17 administrative regions in the 2027–2028 phase. Findings must reach DOH leadership and USAID Philippines before that budget window opens, not after. A second audience is the broader multilateral community — WHO WPRO, the World Bank, and Gavi — for whom iDEWS represents a test case for community-anchored digital surveillance as a pandemic preparedness model. That audience operates on a longer horizon and needs the full methodological record, not just the headline result.

8.2 Dissemination Audiences and Formats

Table 3: Dissemination audiences, decisions, and matched formats.

Audience	Decision / Interest	Format
Philippines DOH & USAID Philippines	Scale-up decision; portfolio reallocation	Executive brief and presentation timed to FY 2027–28 budget cycle
Regional Health Offices and LGU health officers	Operational lessons; targeting and deployment refinements	Implementation findings report; regional debriefs
WHO WPRO and multilateral partners	Pandemic preparedness model; replicability across Southeast Asia	Full evaluation report; working paper for field dissemination
Academic and policy research community	Methodological contribution; PSM in surveillance program evaluation	Peer-reviewed article; public pre-analysis plan and results

8.3 Timeline and Milestones

The evaluation schedule is built backward from a fixed external deadline — the DOH’s FY 2027–28 budget development — so that findings arrive in time to inform the scale-up decision they are meant to inform rather than after it. The evaluation is resourced within the program’s \$9.6M envelope, with the evaluation team’s data collection and analysis costs forming a minor share of that total

relative to the program’s deployment and training activities. Table 4 maps the key milestones to contract month.

An interim report at approximately month 18 delivers early implementation findings — platform adoption rates, reporting completeness, and alert accuracy — giving DOH and USAID an operational read before the 24-month outcome data matures. The full evaluation report, incorporating matched-comparison outcome estimates at 12 and 24 months, is delivered by approximately March 2027, timed to the DOH budget development cycle. Field dissemination and any peer-reviewed publication follow on a research timeline, unconstrained by the budget deadline.

Table 4: Evaluation milestones by approximate contract month.

Phase	Months	Key activities and deliverables
Inception	1–3	Finalize design; register matching and analysis specification before outcomes are observed; establish CCHAIN/PIDSR barangay linkage.
Baseline	3–6	Assemble pre-deployment covariate and incidence data; estimate propensity scores; construct matched sample.
Monitoring	6–24	Quarterly performance monitoring against review thresholds; ML alert reconciliation per outbreak event.
Interim report	~18	Early implementation findings: adoption, reporting completeness, alert accuracy.
Outcome waves	12, 24	Matched-comparison outcome estimates on incidence, mortality, and case fatality.
Final report	~Mar 2027	Full evaluation report and executive brief, timed to the FY 2027–28 budget window.

8.4 Pathways to Policy Uptake

The most direct pathway to uptake is the budget cycle — findings timed to the scale-up decision carry weight that findings delivered afterward do not. The second pathway is operational: Regional Health Offices and LGU health officers can act on implementation findings to improve platform adoption and alert response regardless of whether a formal scale-up is funded, so that even a constrained budget outcome absorbs the pilot’s lessons. The plan also anticipates honest communication of limits — a null result on incidence or mortality does not mean iDEWS failed; it may mean the binding constraint is response capacity rather than detection, and that reframing is itself actionable intelligence for the next phase of investment.

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